Assignment 5

UID - 2026800018
Name - Shreya Balasaheb Bhoite

```
/*
*@breif To check whether an username is valid according to the conditions.
*
*Conditions
* 1. Length should be minimum 5 and maximum 15
* 2. Username should contain only alphabets and digits (no spaces or special characters)
*
* This code also prints processed username by converting all alphabets to lowerCase.
*
* The code prints username validity or invalidity at the end.
*
*@author Shreya Bhoite
*@date 28-10-2026
*
*/

#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main()
{
   //Declaring a char array to store username
   char username[100];

   //Taking username input from user
   printf("Enter a username: ");
   fgets(username,sizeof(username),stdin);

   //Calculating length of the username input
   int length = strlen(username);

   int valid = 0;          //@param valid Variable to check whether username is valid or not


   //Checking if any characters other than alphanumerics are present
   for (int i=0;username[i] != '\0';i++) {
```

```
        if (isalnum((unsigned char)username[i]) == 1) {
            valid = 1;
            break;
        }
    }


    //Converting entire string to lowerCase
    for (int i=0;username[i] != '\0';i++) {
        username[i] = tolower((unsigned char)username[i]);
    }

    //Printing processed username
    printf("Processed username: %s",username);


    //Checking if all the conditions are true and printing output
    if (length >= 6 && length <=15 && valid == 0) {
        printf("Username is valid");
    } else {
        printf("Username is invalid");
    }

    return 0;
}
```

```
Enter a username: govind@gmail.com
Processed username: govind@gmail.com
Username is invalid
```

```
Enter a username: Govind007
Processed username: govind007
Username is invalid
```

```
Enter a username: govindisaProfessorofpractice
Processed username: govindisaprofessorofpractice
Username is invalid
```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>


int checkLength(char str[])
{
   int length = strlen(str);

   return length;
}

int isStrong(char str[])
{
   int lengthOfString = strlen(str);
   int upperCase = 0;
   int lowerCase = 0;
   int digit = 0;

   int i=0;

   while (i<lengthOfString && (upperCase == 0 || lowerCase == 0 || digit == 0) )
   {
      if (isupper(str[i]) != 0) {
         upperCase++;
      } else if (islower(str[i]) != 0) {
         lowerCase++;
      } else if (isdigit(str[i]) != 0) {
         digit++;
      }
      i++;
   }

   if (upperCase != 0 && lowerCase != 0 && digit != 0) {
      printf("\nStrong Password");
   } else {
      printf("\nWeak Password (must contain uppercase, lowercase and digit)");
   }

   return 0;
}
```

```
void toLowerCase(char str[])
{
    for (int i=0;i<strlen(str);i++) {
        str[i] = tolower(str[i]);
    }

    printf("\nPassword in lowercase: %s",str);
}


int main()
{
    char password[100];

    printf("Enter a password: ");
    scanf("%s",password);

    printf("\nPassword Length = %d",checkLength(password));
    isStrong(password);

    toLowerCase(password);
}
```

```
Enter a password: govind123

Password Length = 9
Weak Password (must contain uppercase, lowercase and digit)
Password in lowercase: govind123
```

```
Enter a password: Govind123

Password Length = 9
Strong Password
Password in lowercase: govind123
```